If $\bar{x}$ is the sample mean and μ is the population mean of the characteristic X then the sampling error is $\bar{x} - \mu$. The sampling error may be positive or negative or zero.

2.9.2 Non-Sampling Errors

The non-sampling errors arise due to various causes right from the beginning stage when the survey is planned and designed to the final stage where the data are processed and analyzed. Non sampling errors are more serious than the sampling errors because a sampling error can be minimized by taking a large sample. It is difficult to minimize non sampling errors, even if a large sample is taken. The main sources of non-sampling errors are now described.

(i) Non-response:

The errors due to non-response may occur due to omission or lapse on the part of the interviewer, or the refusal on the part of the respondents to questions or because of the non-availability of the individuals during the period of survey.

(ii) Errors in measurement:

The measuring device may be biased or inaccurate. The respondent may not know the correct answer and may give imprecise answers. Common examples are questions on age, income, and events that happened in the past. The interviewer may also fail to record the responses correctly. Errors in measurement include errors in coding, editing, and tabulation.

Coverage errors:

The coverage errors are classified as **'under coverage errors'** and **'over coverage errors'.** Under-coverage errors occur in the following situations:

- The selected unit in the sampling frame is not interviewed by the investigator.
- The selected unit is incorrectly classified as ineligible for surveys
- The unit is omitted or skipped by the interviewer.

Similarly, over-coverage occurs under the following situations:

- The sampling frame covers ineligible units.
- The frame may contain the same unit more than once.

The errors cannot be ignored since the cumulative effect of these errors affect the objectives of the survey.

Organising a sample survey

The above said things provide a comprehensive idea about collection of data. However, when one decides to collect data through sampling the following steps are to be followed.

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